

PAPER • OPEN ACCESS

Heterogeneity of Tectonic Regimes in Sulawesi Island: Applying Unsupervised Machine Learning to Focal Mechanism Data

To cite this article: Haidir Jibran and Muhammad Taufiq Rafie 2025 *IOP Conf. Ser.: Earth Environ. Sci.* **1525** 012002

View the [article online](#) for updates and enhancements.

You may also like

- [The Role of Geological Information's in The Development of Geopark Area in South Sulawesi, Indonesia](#)
Hendra Pachri, Busthan Azikin and Ilham Alimuddin
- [Morphometrics of two *Tylomelania* gastropods from the Pattunuang River, Maros, South Sulawesi](#)
Magdalena Litaay, Like Ayu Sutrisno, Andi Aliah Hidayani et al.
- [Checklist of endemic \(*Adrianichthyidae*, *Gobiidae*, *Zenarchopteridae*\) and introduced fish in Lake Poso, Sulawesi, Indonesia](#)
K P Bandjolu, E Madiyono, M Herjayanto et al.

Heterogeneity of Tectonic Regimes in Sulawesi Island: Applying Unsupervised Machine Learning to Focal Mechanism Data

Haidir Jibran¹ and Muhammad Taufiq Rafie^{2*}

¹Study Program of Geophysics, Geophysics Department, Faculty of Mathematics and Natural Sciences, Hasanuddin University, 90245, Indonesia

²Solid Geophysics Laboratory, Faculty of Mathematics and Natural Sciences, Hasanuddin University, 90245, Indonesia

*E-mail: taufiqrafie@unhas.ac.id

Abstract. Sulawesi Island was formed in the tripel junction area between the Sunda plate, Australian plate, and Philippine Sea plate. The collision between the three plates resulted in Sulawesi island having a variety of fault mechanisms. To better understand the tectonic processes in Sulawesi, we clustered earthquake events in the region, which were then analyzed based on their focal mechanisms. The data we used is the distribution of focal mechanisms around the island of Sulawesi for 1390 events from 1967 to 2022 that we obtained from a combination of the ISC GEM, Global CMT, and BMKG earthquake catalogs. We applied one of the unsupervised machine learning technique that is Density-Based Spatial Clustering of Application with Noise (DBSCAN) to cluster these earthquake events. This method clusters earthquake events based on the density of the events using two parameters: the minimum number of events that occur in cluster (minPts), and distance radius for neighbouring points (ϵ). Furthermore, we analysed the density of the focal mechanism in each cluster to comprehend its fault mechanism heterogeneity. We obtained ten earthquake clusters, using the parameters minPts=15 and $\epsilon=0.40$, where these parameters explain the reliability of the tectonic setting in Sulawesi. Based on the clusterization, we infer that seven clusters have compressional regimes located in the northern arm of Sulawesi, the Molucca Sea Double Subduction area, the West Sangihe trench, the North-Vergent thrust, and the Nusa Tenggara Back Arc thrust. Additionally, we found that the clusters associated with the Palu-Koro Fault and Matano Fault exhibit characteristics of a transform regime, while the cluster located between the Lalanga and Tongian ridges is characterized by an extensional setting.

Keywords: Celebes, DBSCAN, Tectonic Regime

1. Introduction

Sulawesi Island has tectonic activity that reflects the complexity of its tectonic formation [1]. The "Latest Miocene Collision" that formed the island of Sulawesi triggered significant deformation of the region, including initiating rotational movement of the island's microblocks [2]. As a result of



the movement of these microblocks, and the subduction of the Celebes Sea along the North Sulawesi trench with the Central Sulawesi Fault System [3] the island's seismicity activity has a high degree of heterogeneity.

As a step towards understanding the tectonic processes in a region, a study of the kinematics and stress fields associated with the seismicity of the region is necessary [4]. However, one of the basic assumptions in conducting these studies is the homogeneity of the region to be studied. To accommodate this prerequisite, we clustered earthquake events that occurred in the area around the island of Sulawesi based on their similar seismicity. Through these clusters, we resolved what they imply to the heterogeneous tectonic regimes in Sulawesi Island.

2. Data and Methodology

In this study, we used focal mechanism catalogues occurring from April 1967 to september 2022, with $4.0 < M_w < 7.0$ and focal depth ≤ 50 km, compiled by Global Centroid Moment Tensor (CMT) [5,6], International Seismological Centre – Global Earthquake Model (ISC-GEM) [7,8,9], and BMKG [10]. From these compilation, we obtained 1390 focal data (Figure 1) which will be used for the clustering process.

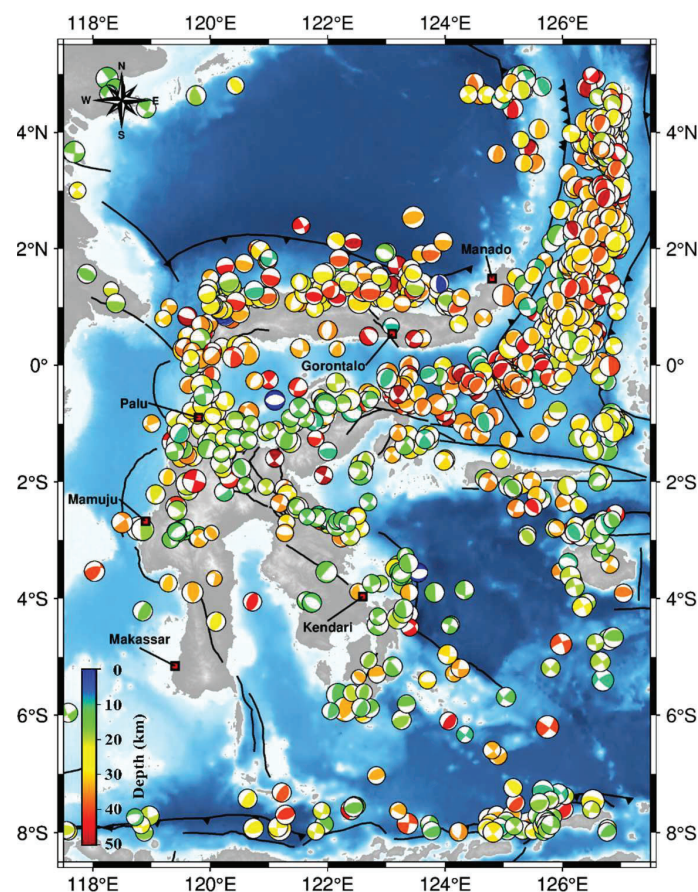


Figure 1. Map showing the distribution of focal mechanisms data, with $4.0 < M_w < 7.0$ and depth ≤ 50 km. The surface fault traces around Sulawesi Island were obtained from PUSGEN (2017) [11]. The red box indicates the major city with highly dense population.

We first clustered the data using Density-Based Spatial Clustering of Application with Noise (DBSCAN) [12], an unsupervised machine learning method, to identify clusters where the focal data originated from uniform principal mechanisms, corresponding to relatively homogeneous areas. DBSCAN forms clusters based on data density, using two parameters: the minimum number of points required for a cluster (minPts) and the distance radius for neighboring points (ε) [13]. This method is widely used in areas with a pronounced spatially clustered seismic activity [13,14].

To optimize the clustering parameters, we experimented with various combinations of minPts and ε . We used a minPts value of 20, because this value is utilized as the minimum number of events for testing the iterative joint inversion algorithm [15]. Additionally, we explored minPts values of 15 and 10 to assess the variation in cluster characteristics according to the tectonic setting. Each minPts value was paired with ε values ranging from 0.05 to 0.50, in increments of 0.05 (Table 1). We then applied ternary diagrams and plotted the P-/T- axis distributions for each cluster generated by different parameter pairs to determine which parameter combinations best represented the distinctive characteristics of each tectonic setting.

Table 1. Number of clusters formed for each pair of parameters

Parameters	Number of Cluster	Parameters	Number of Cluster	Parameters	Number of Cluster
minPts10 ε 0.05	3	minPts15 ε 0.05	0	minPts20 ε 0.05	0
minPts10 ε 0.10	9	minPts15 ε 0.10	6	minPts20 ε 0.10	2
minPts10 ε 0.15	8	minPts15 ε 0.15	5	minPts20 ε 0.15	5
minPts10 ε 0.20	12	minPts15 ε 0.20	6	minPts20 ε 0.20	6
minPts10 ε 0.25	15	minPts15 ε 0.25	7	minPts20 ε 0.25	6
minPts10 ε 0.30	16	minPts15 ε 0.30	9	minPts20 ε 0.30	6
minPts10 ε 0.35	11	minPts15 ε 0.35	10	minPts20 ε 0.35	6
minPts10 ε 0.40	11	minPts15 ε 0.40	10	minPts20 ε 0.40	5
minPts10 ε 0.45	11	minPts15 ε 0.45	11	minPts20 ε 0.45	7
minPts10 ε 0.50	8	minPts15 ε 0.50	10	minPts20 ε 0.50	7

Table 2. Distribution of data in each cluster generated with parameters minPts=15 and ε =0.40

Cluster ID	Number of Data	Cluster ID	Number of Data
1	49	6	19
2	81	7	18
3	62	8	742
4	16	9	30
5	24	10	33

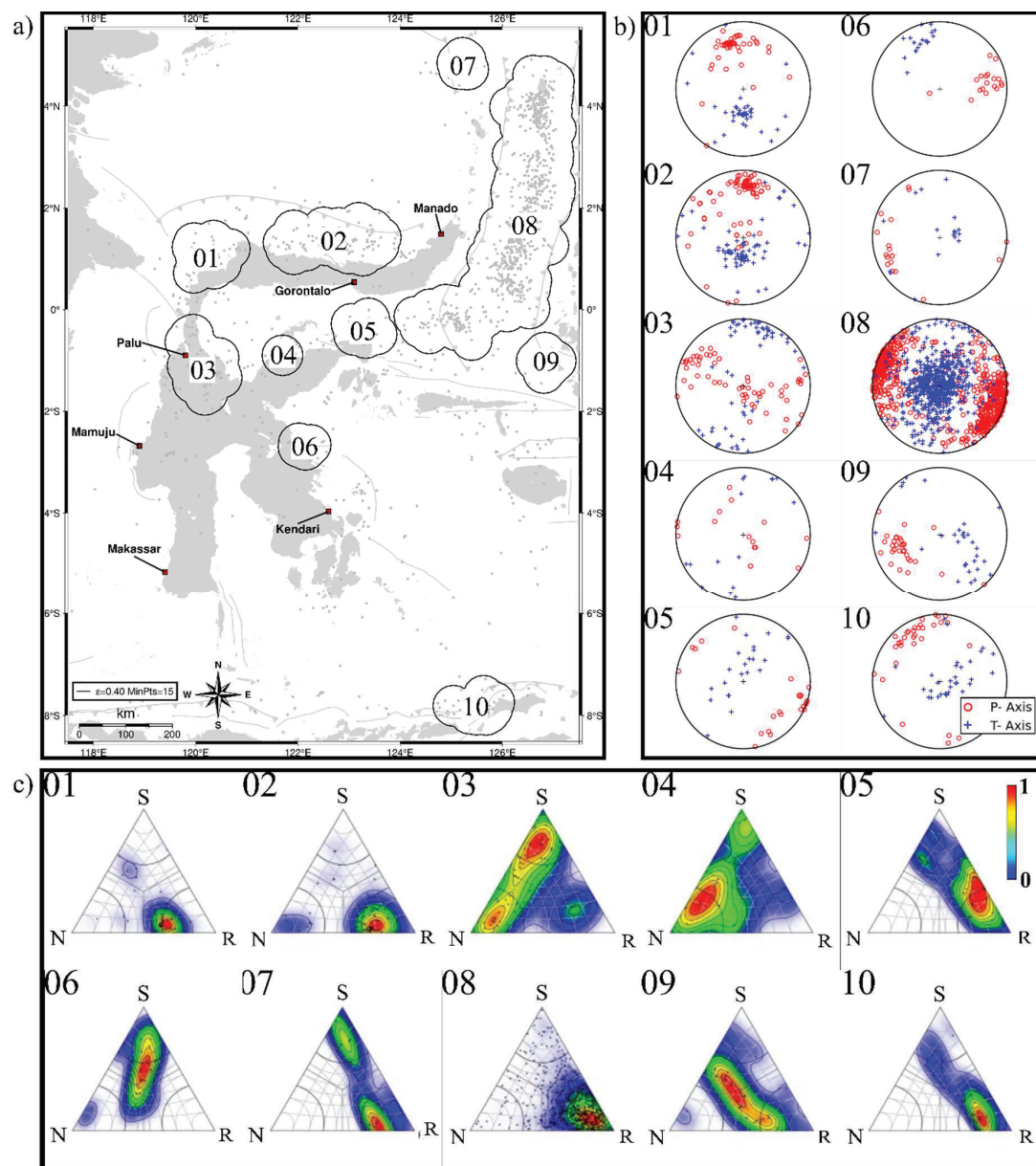


Figure 2. Best clustering results generated using the DBSCAN algorithm with a parameter of minPts=15 and $\epsilon=0.40$. a) The map shows boundaries of each cluster, with red boxes indicating large cities with high population density. b) The distribution of the P-/T- axes within each cluster is illustrated, where red circles indicate the P-axis direction and blue plus represent the T-axis direction. c) A ternary diagram shows the distribution of event mechanisms in each cluster: S represents strike-slip, N indicates normal, and R denotes reverse faulting.

3. Result and Discussion

In this study, the optimal parameters for the DBSCAN algorithm were determined to be minPts=15 and $\epsilon=0.40$. These parameters were chosen based on the distribution patterns of focal mechanism

types within each cluster, as represented in ternary diagrams and the P-/T- axis distribution. Using these settings, the algorithm successfully clustered the seismic events into 10 clusters (Figure 2), each containing a different number of events (Table 2). Of these clusters, six were located on the island of Sulawesi, while the remaining four were distributed around the island. In addition to the clustered earthquake events, we identified 315 earthquake events categorized as noise, which are evenly distributed across the study area. These events were excluded from the stress inversion analysis due to their insufficient density based on the defined parameters. This exclusion ensures that the data used for stress inversion consist solely of earthquakes generated from a uniform stress orientation, thereby preventing the inclusion of events influenced by different stress regimes. By doing so, potential uncertainties in the analysis are minimized.

Clusters 01 and 02 consist of 49 and 81 earthquake events, respectively, distributed across the northern part of the northern arm of Sulawesi. Both clusters exhibit a compressional regime, characterized by a predominance of reverse fault mechanisms, resulting from the subduction activity of the Sulawesi Sea in the region. Cluster 03, which contains 62 events, displays a more diverse range of focal mechanisms. The ternary diagram reveals two dominant earthquake types within this cluster: strike-slip and normal. The primary fault in this cluster is the Palu-Koro Fault, which exhibits a transform regime characterized by left-lateral movement, resulting in events with strike-slip mechanisms. Additionally, normal mechanisms are generated by smaller faults in the area, including the Palolo Fault and Bada Fault [11]. Clusters 04 and 05, containing 16 and 24 events respectively, exhibit complex seismic characteristics due to their location in regions where the latest Miocene collision occurred [1]. According to the ternary diagram, Cluster 04 is dominated by normal faulting, indicating an extensional setting influenced by the behavior of the Lalanga ridge and Tongian ridge [1]. In contrast with Cluster 04, Cluster 05 is dominated by reverse fault mechanisms, which are attributed to its position within the North-Vergent thrust zone [1]. Cluster 06 consists of earthquakes occurring around the Matano Fault. The ternary diagram for this cluster indicates a predominance of strike-slip mechanisms, resulting from the shear faulting activity along the Matano Fault. The remaining four clusters, located outside the mainland of Sulawesi Island, are situated in subduction zones and therefore experience compressional regimes. Cluster 07 is positioned in the West Sangihe trench, Clusters 08 and 09 are found within the Molucca Sea Double Subduction, and Cluster 10 is located in Nusa Tenggara Back Arc thrust. [11].

4. Conclusion

Based on the results of this research, the application of unsupervised machine learning using the DBSCAN algorithm can be used effectively in grouping earthquake events in the Sulawesi region according to the similarities in their generating mechanisms. Utilizing the parameters $\text{minPts}=15$ and $\epsilon=0.40$ as the optimized parameter, we identified 10 distinct clusters. Notably, the clusters associated with the Palu-Koro Fault and Matano Fault exhibit a transform regime, while areas between the Lalanga and Tongian ridges are characterized by extensional setting. The remaining seven clusters display compressional regimes, predominantly formed in subduction zones, including those in the northern arm of Sulawesi, the Molucca Sea Double Subduction area, the West Sangihe trench, the North-Vergent thrust, and the Nusa Tenggara Back Arc thrust.

References

- [1] Serhalawan Y and Chen P F 2024 Seismotectonics of Sulawesi, Indonesia *Tectonophysics* **883** 230366 <https://doi.org/10.1016/j.tecto.2024.230366>.
- [2] Socquet A, Simons W, Vigny C, McCaffrey R, Subarya C, Sarsito D, Ambrosius B and Spakman W 2006 Microblock rotations and fault coupling in SE Asia triple junction (Sulawesi, Indonesia) from GPS and earthquake slip vector data *Journal of Geophysical Research: Solid Earth* **111** B8 <https://doi.org/10.1029/2005JB003963>.
- [3] Walpersdorf A, Rangin C and Vigny C 1998 GPS compared to long-term geologic motion of the north arm of Sulawesi *Earth and Planetary Science Letters* **159** (1-2) 47-55 [https://doi.org/10.1016/S0012-821X\(98\)00056-9](https://doi.org/10.1016/S0012-821X(98)00056-9).
- [4] Scarfi L, Langer H, Messina A and Musumeci C 2021 Tectonic regimes inferred from clustering of focal mechanisms and their distribution in space: application to the Central Mediterranean Area *Journal of Geophysical Research: Solid Earth* **126** (1) 2020JB020519 <https://doi.org/10.1029/2020JB020519>.
- [5] Dziewonski A M, Chou T A and Woodhouse J H 1981 Determination of earthquake source parameters from waveform data for studies of global and regional seismicity *Journal of Geophysical Research: Solid Earth* **86** (B4) 2825-2852 <https://doi.org/10.1029/JB086iB04p02825>.
- [6] Ekström G, Nettles M and Dziewoński A M 2012 The global CMT project 2004–2010: Centroid-moment tensors for 13,017 earthquakes *Physics of the Earth and Planetary Interiors* **200** 1-9 <https://doi.org/10.1016/j.pepi.2012.04.002>.
- [7] Storchak D A, Di Giacomo D, Bondár I, Engdahl E R, Harris J, Lee W H, Villaseñor A and Bormann P 2013 Public release of the ISC–GEM global instrumental earthquake catalogue (1900–2009) *Seismological Research Letters* **84** (5) 810-815 <https://doi.org/10.1785/0220130034>.
- [8] Storchak D A, Di Giacomo D, Engdahl E R, Harris J, Bondár I, Lee W H, Bormann P and Villaseñor A 2015 The ISC–GEM global instrumental earthquake catalogue (1900–2009): introduction *Physics of the Earth and Planetary Interiors* **239** 48-63 <https://doi.org/10.1016/j.pepi.2014.06.009>.
- [9] Di Giacomo D, Engdahl E R and Storchak DA 2018 The ISC–GEM earthquake catalogue (1904–2014): status after the extension project *Earth System Science Data* **10** (4) 1877-1899 <https://doi.org/10.5194/essd-10-1877-2018>.
- [10] Massinai M F I, Lantu, Aswad S and Massinai M A 2015 Tectonics earthquake distribution pattern analysis based focal mechanisms (Case study Sulawesi Island, 1993–2012) In *AIP Conference Proceedings* **1658** 1 <https://doi.org/10.1063/1.4915021>.
- [11] National Center for Earthquake Studies of Indonesia (PUSGEN) 2017 *Indonesia's Earthquake Source and Hazard Map* (in Bahasa Indonesia) Indonesia Research and Development Agency of Ministry of Public Work and Housing ISBN 978–602–5489–01–3.
- [12] Ester M, Kriegel H P, Sander J and Xu X 1996 A density-based algorithm for discovering clusters in large spatial databases with noise In *kdd* **96** 34 226-231.
- [13] Wollin C, Bohnhoff M, Vavryčuk V and Martínez-Garzón P 2019 Stress inversion of regional seismicity in the Sea of Marmara Region, Turkey *Pure and Applied Geophysics* **176** 3 1269-1291 <https://doi.org/10.1007/s00024-018-1971-1>.
- [14] Schoenball M and Ellsworth W L 2017 A systematic assessment of the spatiotemporal evolution of fault activation through induced seismicity in Oklahoma and southern Kansas *Journal of Geophysical Research: Solid Earth* **122** 12 10-189 <https://doi.org/10.1002/2017JB014850>.
- [15] Vavryčuk V 2014 Iterative joint inversion for stress and fault orientations from focal mechanisms *Geophysical Journal International* **199** 1 69-77 <https://doi.org/10.1093/gji/ggu224>.